

Bulan 7: Advanced Patterns & Scalability

Minggu 1: Design Patterns & Architecture

Hari 1: Repository Pattern

Topik: Repository pattern, data access abstraction

Proyek: RepositoryPattern

Implement repository pattern

- Create repository interface
- ProductRepository class
- Abstract database operations
- Swap implementations
- Unit test dengan mock repositories
- Teknologi: Design patterns, abstraction

Hari 2: Service Layer Pattern

Topik: Business logic separation, service classes

Proyek: ServiceLayer

Service layer architecture

- Separate business logic dari views
- UserService class
- OrderService dengan complex logic
- Transaction handling
- Reusable business logic
- Teknologi: Service layer pattern

Hari 3: CQRS Pattern Basics

Topik: Command Query Responsibility Segregation

Proyek: CQRSImplementation

Basic CQRS pattern

- Separate read & write models
- Command handlers

- Query handlers
- Different optimizations for reads vs writes
- Event sourcing intro
- Teknologi: CQRS pattern

Hari 4: Dependency Injection

Topik: DI pattern, injector, loose coupling

Proyek: DependencyInjection

Implement dependency injection

- Create service interfaces
- Inject dependencies into views
- django-injector library
- Better testability
- Loose coupling
- Teknologi: Dependency injection

Hari 5: Factory Pattern

Topik: Factory pattern, object creation abstraction

Proyek: FactoryPattern

Factory pattern untuk models

- NotificationFactory
- Different notification types (Email, SMS, Push)
- Strategy pattern integration
- Polymorphic creation
- Teknologi: Factory pattern

Hari 6: Observer Pattern & Django Signals

Topik: Observer pattern dengan signals, event-driven

Proyek: EventDrivenArchitecture

Event-driven system

- Custom events

- Multiple observers (signals)
- Async event handling
- Event bus simulation
- Decouple components
- Teknologi: Observer pattern, signals

Hari 7: Review & Mini Project

Mini Project: CleanArchitectureAPI

Clean architecture implementation

Layers:

- Domain layer (business entities)
- Use cases layer (business logic)
- Interface adapters (serializers, views)
- Infrastructure (database, external services)

Features:

- Order management system
- Repository pattern untuk data access
- Service layer untuk business logic
- Dependency injection
- CQRS untuk orders (read/write separation)
- Event-driven (order events)
- Unit tests dengan mocks
- Integration tests
- Teknologi: Clean architecture, Design patterns, DI

Minggu 2: Caching Strategies & Performance

Hari 8: Redis Deep Dive

Topik: Redis data structures, commands, use cases

Proyek: RedisExploration

Advanced Redis usage

- Connect to Redis

- String, List, Set, Hash operations
- Expire keys
- Pub/Sub basics
- Redis as session store
- Teknologi: Redis

Hari 9: Cache Strategies

Topik: Cache-aside, write-through, cache warming

Proyek: CacheStrategies

Implement caching patterns

- Cache-aside pattern
- Cache warming on startup
- Cache invalidation strategies
- TTL management
- Cache key naming conventions
- Teknologi: Caching patterns

Hari 10: Query Result Caching

Topik: Cache QuerySets, cache expensive queries

Proyek: QueryCache

Cache database queries

- Cache expensive aggregations
- Cache search results
- Invalidate on model save
- Cache user-specific data
- Cache statistics
- Teknologi: Query caching

Hari 11: API Response Caching

Topik: Cache API responses, ETags, cache headers

Proyek: APICaching

Cache REST API responses

- @cache_page for views
- Cache per user
- Vary header
- ETags implementation
- Conditional requests
- Teknologi: API caching, HTTP caching

Hari 12: Database Connection Pooling

Topik: PgBouncer, connection pooling, database optimization

Proyek: ConnectionPooling

Optimize database connections

- Setup PgBouncer (simulate)
- Configure pool size
- Monitor connections
- Connection timeout
- Transaction pooling vs session pooling
- Teknologi: Connection pooling

Hari 13: Database Indexing Strategy

Topik: Advanced indexing, composite indexes, partial indexes

Proyek: IndexingStrategy

Optimize with indexes

- Analyze slow queries
- Create appropriate indexes
- Composite indexes for complex queries
- Partial indexes
- Index maintenance
- EXPLAIN ANALYZE
- Teknologi: Database indexing

Hari 14: Review & Mini Project

Mini Project: HighPerformanceAPI

High-performance e-commerce API

Features:

- Product catalog (100,000+ products)
- Redis caching layer
- Cache warming untuk popular products
- Cached search results (expire 5 min)
- Cached aggregations (categories count)
- Database connection pooling
- Optimized indexes
- Query optimization (select_related, prefetch_related)
- API response caching dengan ETags
- Cache invalidation on updates
- Performance benchmarks
- Load testing dengan locust
- Response time < 100ms untuk cached
- Teknologi: Redis, Caching, Indexing, Optimization

Minggu 3: Horizontal Scaling & Load Balancing

Hari 15: Stateless Application Design

Topik: Stateless vs stateful, session management

Proyek: StatelessAPI

Design stateless application

- Remove local state
- Session in Redis
- JWT tokens (stateless auth)
- File storage in S3 (simulate)
- No server affinity required
- Teknologi: Stateless design

Hari 16: Load Balancer Basics

Topik: Load balancing algorithms, Nginx load balancer

Proyek: LoadBalancer

Setup load balancer

- Nginx as load balancer
- Round-robin algorithm
- Multiple backend servers
- Health checks
- Upstream configuration
- Teknologi: Load balancing

Hari 17: Session Management at Scale

Topik: Distributed sessions, Redis session backend

Proyek: DistributedSessions

Shared session storage

- Redis as session backend
- Configure django-redis-sessions
- Session expiry
- Session data optimization
- Concurrent session handling
- Teknologi: Distributed sessions

Hari 18: Database Read Replicas

Topik: Master-slave replication, read replicas, routing

Proyek: DatabaseReplication

Setup read replicas

- Configure database router
- Read from replica
- Write to master
- Lag handling
- Failover simulation
- Teknologi: Database replication

Hari 19: Message Queues & Task Distribution

Topik: RabbitMQ, task distribution, worker scaling

Proyek: DistributedTasks

Distribute tasks across workers

- RabbitMQ as broker (simulate)
- Multiple Celery workers
- Task routing
- Priority queues
- Worker autoscaling
- Teknologi: Message queues, Celery

Hari 20: Microservices Communication

Topik: Service-to-service communication, API gateway

Proyek: MicroservicesCommunication

Inter-service communication

- HTTP-based communication
- Service discovery (simulate)
- Circuit breaker pattern
- Retry mechanism
- Timeout handling
- Teknologi: Microservices patterns

Hari 21: Review & Mini Project

Mini Project: ScalableArchitecture

Horizontally scalable application

Architecture:

- Load balancer (Nginx)
- 3x Application servers (Django)
- Redis cluster (cache & sessions)
- PostgreSQL master + 2 read replicas
- Celery workers (3+ instances)

- Message queue (RabbitMQ)

Features:

- Stateless application design
- JWT authentication
- Session in Redis
- Media files in S3 (simulate)
- Database read/write splitting
- Distributed caching
- Task distribution
- Health checks
- Auto-scaling simulation
- Monitoring all components
- Teknologi: Load balancing, Replication, Redis cluster, Microservices

Minggu 4: Advanced Topics & Best Practices

Hari 22: GraphQL with Django

Topik: GraphQL basics, graphene-django

Proyek: GraphQLAPI

GraphQL API dengan Django

- Install graphene-django
- Define schema
- Queries & mutations
- Nested queries
- Query optimization (N+1 problem)
- GraphQL interface
- Teknologi: GraphQL, graphene-django

Hari 23: Elasticsearch Integration

Topik: Full-text search, Elasticsearch, django-elasticsearch-dsl

Proyek: SearchEngine

Advanced search dengan Elasticsearch

- Setup Elasticsearch (simulate)
- Index Django models
- Full-text search
- Fuzzy search
- Faceted search
- Search suggestions
- Teknologi: Elasticsearch

Hari 24: Rate Limiting & Throttling

Topik: Advanced rate limiting, distributed rate limiting

Proyek: AdvancedRateLimiting

Comprehensive rate limiting

- DRF throttling classes
- Redis-based rate limiting
- Per-user limits
- Per-endpoint limits
- Burst allowance
- Rate limit headers
- Teknologi: Rate limiting, Redis

Hari 25: API Gateway Pattern

Topik: API gateway, Kong, request routing

Proyek: APIGateway

Implement API gateway

- Kong as API gateway (simulate)
- Request routing
- Authentication at gateway
- Rate limiting at gateway
- Request transformation
- Teknologi: API gateway

Hari 26: Multi-tenancy

Topik: Multi-tenant architecture, tenant isolation

Proyek: MultiTenantApp

Multi-tenant SaaS application

- Tenant model
- Tenant middleware
- Schema-based tenancy
- Tenant-specific data isolation
- Subdomain routing
- Tenant database routing
- Teknologi: Multi-tenancy

Hari 27: Serverless Functions

Topik: AWS Lambda, serverless Django, Zappa

Proyek: ServerlessAPI

Django on serverless

- Zappa setup (simulate)
- Deploy Django to Lambda
- API Gateway integration
- Cold start optimization
- Serverless benefits & limitations
- Teknologi: Serverless, AWS Lambda

Hari 28: Review & Final Project Bulan 7

Final Project: EnterpriseSaaS

Enterprise SaaS platform

Architecture:

- API Gateway (Kong)
- Load balancer
- Multi-tenant Django application
- GraphQL & REST API

- Elasticsearch for search
- PostgreSQL (master + replicas)
- Redis cluster (cache, sessions, rate limiting)
- Celery workers
- S3 for file storage

Features:

- Multi-tenant system (subdomain-based)
- Tenant isolation (schema per tenant)
- GraphQL API untuk complex queries
- REST API untuk simple CRUD
- Advanced search dengan Elasticsearch
- Rate limiting per tenant
- Background job processing
- Real-time notifications (WebSocket)
- File upload & processing
- Analytics & reporting
- Audit logs per tenant
- Tenant admin dashboard
- Horizontal scaling ready
- Comprehensive monitoring
- Auto-scaling policies

Security:

- Tenant data isolation
- Role-based access control (RBAC)
- API authentication (JWT)
- Rate limiting
- SQL injection protection
- XSS protection
- CSRF protection

Performance:

- Response time < 200ms (p95)
- Support 10,000+ concurrent users
- Database query optimization
- Caching strategy
- CDN for static files

Teknologi: Multi-tenancy, GraphQL, Elasticsearch, Redis cluster, API Gateway, Horizontal scaling, Monitoring

Progress Check Bulan 7:  Design patterns (Repository, Service layer, CQRS, DI, Factory, Observer) 

Clean architecture  Advanced caching strategies  Redis deep dive  Database optimization (indexing, connection pooling)  Stateless design  Load balancing  Database replication  Horizontal scaling  Microservices patterns  GraphQL  Elasticsearch  Advanced rate limiting  Multi-tenancy 
API gateway

Total Proyek: 60+ proyek & 4 major architecture projects **Siap lanjut:** Bulan 8 - Real-world Projects & Portfolio